TERNARY FORCING ON THE HEXAGONAL LATTICE: EISENSTEIN INTEGERS, CASCADE MODULI, AND A CONJECTURED LEVEL OF DISTRIBUTION 5/8

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ABSTRACT. We discuss primes in arithmetic progressions to the moduli $q = 3^K$ ("cascade moduli") and record the arithmetic of the Eisenstein integers $\mathbb{Z}[\omega]$ at the ramified prime 3. The original version of this paper claimed a level of distribution $\theta = 5/8$ for these moduli. Its main theorem, a Bombieri–Vinogradov-type sum over the moduli $3^K \le x^{5/8-\varepsilon}$, is true, but it remains true with 5/8 replaced by any fixed $\theta < 1$: the family contains only $O(\log x)$ moduli, and the trivial bound together with the Siegel–Walfisz theorem already suffices. It therefore carries no information about 5/8. The meaningful statement, an asymptotic formula for $\psi(x; 3^K, a)$ for each single modulus $3^K \le x^{5/8-\varepsilon}$, is stated here as a conjecture. It would go beyond what the Generalised Riemann Hypothesis gives (moduli up to $x^{1/2}$). For prime-power moduli the known ranges have improved from $x^{3/8}$ (Barban–Linnik–Tshudakov) through $x^{2/5}$ (Gallagher) to $x^{5/12}$ (Huxley), with lower bounds of the correct order up to $x^{0.4736}$ (Banks–Shparlinski). The argument offered for 5/8 in the original version is withdrawn, and Section 7 records where each step fails.

Revision note (November 2026). The title now reads "a Conjectured Level of Distribution 5/8". Theorem 1.1 is restated for every fixed $\theta < 1$ with a correct proof, and the per-modulus level 5/8 is now Conjecture 1.3, which would go beyond GRH. The proof of 5/8 is withdrawn: there is no three-level filtration (the $(1-\omega)$-adic filtration of $\mathbb{Z}[\omega]/(3^K)$ has $2K$ steps), Theorem 1.1 involves no sieve weights, the bound $x^{1/2}Q^{3/4}$ was never derived, the Kloosterman bound $|S(a, b; 3^K)| \ll (3^K)^{3/8+\varepsilon}$ is false, and the final inequality $x^{1/2}Q^{5/8} \le x^{1-\varepsilon}$ would give $Q \le x^{4/5}$, not $x^{5/8}$. The appendix on "conditions W1–W3 of Watt's theorem" is withdrawn, since Watt's theorem is a t-aspect mean value with no such conditions. The uniqueness theorem (former Theorem 5.1) and every claim that Pascadi's work confirms or bounds anything here are withdrawn. The descriptions of the Bombieri–Vinogradov theorem, Polymath8 and Pascadi, the statement of the Weil bound and the bibliography are corrected. The Eisenstein-integer arithmetic, the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$, the inert/split criterion, the Legendre-transform proposition and the factorisation $L(s, \chi \circ \mathrm{N}) = L(s, \chi)L(s, \chi\chi_{-3})$ are kept.

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1. Introduction

1.1. **Background.** The Bombieri–Vinogradov theorem [5, 18] states that for every $A > 0$ there is $B = B(A)$ such that

$$\sum_{q \le x^{1/2}/(\log x)^B} \max_{a:\ \gcd(a,q)=1} \left| \psi(x;q,a) - \frac{x}{\varphi(q)} \right| \ll \frac{x}{(\log x)^A}.$$

It is a statement about the *average* over the moduli $q \le x^{1/2}(\log x)^{-B}$, not about each modulus separately. The primes are said to have *level of distribution* θ if the same bound holds with the sum extended over $q \le x^{\theta-\varepsilon}$, for every $\varepsilon > 0$; Bombieri–Vinogradov gives $\theta = 1/2$. The Generalised Riemann Hypothesis (GRH) also gives level $1/2$ and does not push it beyond $1/2$; the Elliott–Halberstam conjecture, a separate conjecture, asserts every $\theta < 1$.

Levels above $1/2$ are known only for modified problems. Bombieri, Friedlander and Iwaniec [6] obtained $4/7$ for a fixed residue class with well-factorable weights. Zhang [20] and Polymath8a [14] obtained levels slightly above $1/2$ (up to $1/2 + 7/300 \approx 0.5233$ in [14]) for a fixed residue class and squarefree moduli whose prime factors are all at most x^δ (more precisely, densely divisible moduli). Pascadi [13] proved that smooth numbers, and primes weighted by triply-well-factorable weights, are equidistributed in arithmetic progressions to general moduli up to $x^{5/8-o(1)}$, using large-sieve inequalities for exceptional Maass forms together with the Kim–Sarnak bound. None of these results concerns the moduli 3^K, which are not squarefree for $K \ge 2$.

The present paper concerns the sparse family of *cascade moduli* $q = 3^K$. For such a family a summed statement of Bombieri–Vinogradov type is uninformative (Theorem 1.1 and Remark 1.2); the meaningful question is how large 3^K may be while each single modulus still satisfies the prime number theorem in progressions (Conjecture 1.3).

1.2. **The summed estimate.**

Theorem 1.1 (Summed estimate for cascade moduli)**.** *For every fixed $\theta \in (0,1)$ and every $A > 0$,*

$$\sum_{\substack{K \ge 1 \\ 3^K \le x^\theta}} \max_{\substack{a \bmod 3^K \\ \gcd(a,3)=1}} \left| \psi(x;3^K,a) - \frac{x}{\varphi(3^K)} \right| \ll_A \frac{x}{(\log x)^A},$$

where $\psi(x;q,a)=\sum_{\substack{n\le x\\ n\equiv a \pmod q}}\Lambda(n)$.

The original version stated Theorem 1.1 with $\theta=5/8-\varepsilon$ and deduced from it a "level of distribution $\theta_W=5/8$ over cascade moduli". That deduction is withdrawn. The proof is in Section 4.

Remark 1.2 (Why Theorem 1.1 says nothing about 5/8)**.** The family $\{3^K\le x^\theta\}$ has only about $\theta\log x/\log 3$ members, so the trivial bound handles every modulus larger than a power of $\log x$ (Section 4). The argument never uses $\theta<5/8$; in fact it works for every fixed $\theta>0$, including $\theta\ge 1$. A "level" defined as the supremum of the admissible θ is therefore not finite for this family. Numerically, at $x=2\cdot 10^7$ the sum in Theorem 1.1 taken over *all* $3^K\le x$ (that is, up to $\theta\approx 0.98$) equals 10,410, while $x/(\log x)^2\approx 70{,}767$. At the same time the normalised error $\varphi(q)\max_a|\psi(x;q,a)-x/\varphi(q)|/x$ is 0.28 at $q=3^9\approx x^{0.59}$ and exceeds 1 for every $q=3^K$ with $3^{12}\le q\le x$. The summed statement does not see where the individual moduli stop being equidistributed. (The computation is `audit/scripts/p04/thm11_trivial.py` in the programme repository.)

1.3. **The per-modulus question.**

Conjecture 1.3 (Per-modulus level 5/8 for cascade moduli)**.** *For every* $\varepsilon>0$ *and* $A>0$,

$$\max_{\substack{a \bmod 3^K\\ \gcd(a,3)=1}}\left|\psi(x;3^K,a)-\frac{x}{\varphi(3^K)}\right|\ll_{\varepsilon,A}\frac{x}{\varphi(3^K)(\log x)^A},$$

uniformly for all $K\ge 1$ *with* $3^K\le x^{5/8-\varepsilon}$.

Remark 1.4 (Status of Conjecture 1.3)**.** We record three points.

(a) *It goes beyond GRH.* For a single modulus q, GRH gives $\psi(x;q,a)=x/\varphi(q)+O(x^{1/2}\log^2 x)$ [8, Ch. 20], which has the form required in Conjecture 1.3 only when $\varphi(q)\le x^{1/2}(\log x)^{-A-2}$. The conjecture asks for moduli up to $x^{5/8-\varepsilon}$, so it asserts more than GRH is known to give.

(b) *Known results.* For a fixed odd prime p, the asymptotic formula for primes in progressions to the modulus p^N is known uniformly for $p^N\le x^{3/8-\varepsilon}$ (Barban, Linnik and Tshudakov [4]), for $p^N\le x^{2/5-\varepsilon}$ (Gallagher [9]) and for $p^N\le x^{5/12-\varepsilon}$ (Huxley). Banks and Shparlinski [3] obtained lower bounds of the correct order of magnitude up to $x^{0.4736}$. See Baier and Pujahari [1] for this history. All of these exponents are below 1/2.

(c) *Nothing in this paper supports the exponent* 5/8. Section 5 explains where the value came from, and Section 7 lists the steps of the original argument and why they fail. The conjecture is stated at 5/8 only because that is the value the programme proposed.

1.4. **Notation.** Throughout: $\omega=e^{2\pi i/3}$, so $\omega^2+\omega+1=0$. The Eisenstein integers are $\mathbb{Z}[\omega]=\{a+b\omega : a,b\in\mathbb{Z}\}$ with norm $\mathrm{N}(a+b\omega)=a^2-ab+b^2$, and $\mathfrak{p}=(1-\omega)$. We write Λ for the von Mangoldt function, φ for Euler's totient, τ for the divisor function, and $\chi_{-3}=\left(\frac{-3}{\cdot}\right)$ for the non-principal character modulo 3. For a Dirichlet character $\chi \pmod q$, χ^0 denotes the principal character. Implicit constants may depend on A, ε, and absolute numerical constants.

1.5. **Position in the programme.** This paper is p04 of a nine-paper programme: p01 [21], p02 [22], p03 [23], p04 (this paper), p05 [24], p06 [25], p07 [26], p08 [27] and p09 [28]. In the revised programme the formula $\theta_k = (2^k - k)/2^k$ is a postulate [21, 24], not a theorem; the value $\theta_W = 5/8$ for cascade moduli is Conjecture 1.3 of this paper; and Condition W3 is true, with a one-line proof [26] (Remark 3.5).

2. The Eisenstein integers at the prime 3

2.1. Ramification of 3 in $\mathbb{Z}[\omega]$.

Lemma 2.1 (Norm of $1-\omega$)**.** $\mathrm{N}(1-\omega) = 3$.

Proof. $\mathrm{N}(a+b\omega) = a^2 - ab + b^2$ with $a = 1$, $b = -1$: $\mathrm{N}(1-\omega) = 1 + 1 + 1 = 3$. □

Proposition 2.2 (Ramification)**.** *In $\mathbb{Z}[\omega]$, the rational prime 3 ramifies: $3 = u\,(1-\omega)^2$ for a unit $u \in \mathbb{Z}[\omega]^\times$, so $(3) = \mathfrak{p}^2$ with $\mathfrak{p} = (1-\omega)$.*

Proof. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1-\omega)^2$, and $-\omega^2$ is a unit in $\mathbb{Z}[\omega]$. □

Proposition 2.3 (Basic invariants of $\mathbb{Q}(\omega)$)**.** *The field $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$ is imaginary quadratic, hence a CM field. Its ring of integers is $\mathbb{Z}[\omega]$, its discriminant is -3, its class number is 1, and its unit group is $\mu_6 = \{\pm 1, \pm\omega, \pm\omega^2\}$. The prime 3 is the only rational prime that ramifies in $\mathbb{Q}(\omega)$.*

Proof. Since $-3 \equiv 1 \pmod 4$, the ring of integers is $\mathbb{Z}[(1+\sqrt{-3})/2] = \mathbb{Z}[1+\omega] = \mathbb{Z}[\omega]$, of discriminant -3. The Minkowski bound is $(2/\pi)\sqrt{3} \approx 1.10 < 2$, so every ideal class contains an ideal of norm 1, and the class number is 1. A unit $a + b\omega$ satisfies $a^2 - ab + b^2 = 1$, whose six integer solutions give μ_6. A prime ramifies if and only if it divides the discriminant. □

Remark 2.4. The prime 3 is also the conductor of the quadratic character χ_{-3} attached to $\mathbb{Q}(\omega)/\mathbb{Q}$. None of this is special to 3: for instance 7 is the only prime that ramifies in $\mathbb{Q}(\sqrt{-7})$, which also has class number 1. The original remark that this ramification gives the moduli 3^K "a geometric structure that generic squarefree moduli do not" is withdrawn; no such structure was exhibited or used.

Proposition 2.5 (Splitting of primes $\ell \neq 3$)**.** *A rational prime $\ell \neq 3$ is inert in $\mathbb{Z}[\omega]$ if and only if $\ell \equiv 2 \pmod 3$. It splits as $\ell = \pi\bar{\pi}$, with $(\pi) \neq (\bar{\pi})$ primes of norm ℓ, if and only if $\ell \equiv 1 \pmod 3$. In particular 2 is inert.*

Proof. Since $\mathbb{Z}[\omega] \cong \mathbb{Z}[x]/(x^2 + x + 1)$, the prime ℓ splits if $x^2 + x + 1$ has a root modulo ℓ and is inert otherwise (the Dedekind–Kummer criterion; for $\ell \neq 3$ there is no repeated root, as the discriminant of $x^2 + x + 1$ is -3). A root r satisfies $r^3 = 1$ and $r \neq 1$ (as $3 \not\equiv 0 \pmod \ell$), so a root exists if and only if $\mathbb{F}_\ell^\times$ has an element of order 3, that is, if and only if $3 \mid \ell - 1$. See also [7]. □

2.2. The $(1-\omega)$-adic filtration.

Proposition 2.6 ($(1-\omega)$-adic filtration)**.** *The ring $\mathbb{Z}[\omega]/(3^K)$, which has 9^K elements, carries the filtration*

$$\mathbb{Z}[\omega]/(3^K) = \mathfrak{p}^0/\mathfrak{p}^{2K} \supset \mathfrak{p}/\mathfrak{p}^{2K} \supset \mathfrak{p}^2/\mathfrak{p}^{2K} \supset \cdots \supset \mathfrak{p}^{2K}/\mathfrak{p}^{2K} = 0$$

with $2K$ steps, and each successive quotient $\mathfrak{p}^j/\mathfrak{p}^{j+1}$ is isomorphic to $\mathbb{Z}[\omega]/\mathfrak{p} \cong \mathbb{F}_3$.

Proof. Proposition 2.2 gives $(3^K) = \mathfrak{p}^{2K}$. Since $\mathfrak{p}$ is principal, multiplication by $(1-\omega)^j$ induces an isomorphism $\mathbb{Z}[\omega]/\mathfrak{p} \to \mathfrak{p}^j/\mathfrak{p}^{j+1}$, and $\mathbb{Z}[\omega]/\mathfrak{p} \cong \mathbb{F}_3$ because $\mathrm{N}(1-\omega) = 3$. □

Remark 2.7 (No three-level structure)**.** The original version asserted that this filtration has "exactly three independent levels" at the scales $3, 3^2, 3^3$ (that is, $j = 2, 4, 6$). That is false. All $2K$ steps are alike. The even steps $j = 2, 4, \ldots, 2K$ correspond to the K rational levels $3, 3^2, \ldots, 3^K$ (the 3-adic filtration of $\mathbb{Z}/3^K\mathbb{Z}$), and nothing distinguishes $j = 2, 4, 6$: for $K \le 2$ there is no level 3^3, and for $K > 3$ the levels $3^4, \ldots, 3^K$ are just like the first three. Moreover, the questions of Section 1 concern $\mathbb{Z}/3^K\mathbb{Z}$, not the ring $\mathbb{Z}[\omega]/(3^K)$.

3. Characters modulo 3^K

Lemma 3.1 (Cyclic structure)**.** $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ *for all* $K \ge 1$*, and* 2 *is a generator.*

Proof. $\varphi(3^K) = 3^K - 3^{K-1} = 2 \cdot 3^{K-1}$. The powers of 2 modulo 9 are $2, 4, 8, 7, 5, 1$, so 2 is a primitive root modulo 9, and a primitive root modulo p^2, for an odd prime p, is a primitive root modulo every p^K [10, Ch. 4]. For $K = 1$, 2 generates $(\mathbb{Z}/3\mathbb{Z})^*$. □

Remark 3.2. Cyclicity is not special to 3: $(\mathbb{Z}/q\mathbb{Z})^*$ is cyclic exactly when $q = 1, 2, 4, p^k$ or $2p^k$ with p an odd prime [10, Ch. 4], so in particular for every odd prime power. Nor is it a source of extra cancellation in Kloosterman sums: for $q = 3^K$ the Weil exponent $1/2$ is attained (Section 7, item (4)). The original remark that the product structure of $(\mathbb{Z}/q\mathbb{Z})^*$ for squarefree q "obstructs" a cancellation that a single cyclic group enables is withdrawn.

Corollary 3.3 (Real characters modulo 3^K)**.** *For each* $K \ge 1$ *there is exactly one real non-principal Dirichlet character modulo* 3^K*, namely the character induced by* χ_{-3}*. In particular, for* $K \ge 2$ *every primitive character modulo* 3^K *is complex.*

Proof. The character group of a cyclic group of even order is cyclic of the same order, so it has exactly one element of order 2. The character induced by χ_{-3} is real and non-principal, so it is that element. It has conductor 3, so it is not primitive modulo 3^K for $K \ge 2$. □

Proposition 3.4 (Base change to $\mathbb{Q}(\omega)$)**.** *Let* χ *be a Dirichlet character modulo* 3^K*,* $K \ge 1$*, and put* $\Psi_\chi(\mathfrak{a}) = \chi(\mathrm{N}\,\mathfrak{a})$ *for nonzero ideals* $\mathfrak{a}$ *of* $\mathbb{Z}[\omega]$ *(so* $\Psi_\chi(\mathfrak{a}) = 0$ *when* $\mathfrak{p} \mid \mathfrak{a}$*). That is,* $\Psi_\chi = \chi \circ \mathrm{N}$*. Then:*

(i) $\Psi_\chi((n)) = \chi(n)^2$ *for every* $n \in \mathbb{Z}$*;*

(ii) for $\Re s > 1$*,*

$$L(s, \chi \circ \mathrm{N}) := \sum_{\mathfrak{a}} \Psi_\chi(\mathfrak{a})\,(\mathrm{N}\,\mathfrak{a})^{-s} = L(s,\chi)\,L(s,\chi\chi_{-3});$$

(iii) if χ *is primitive,* Ψ_χ *is the trivial character for* $K = 1$*, and its conductor is* $\mathfrak{p}^{2K-1}$ *for* $K \ge 2$*.*

Proof. (i) $\mathrm{N}((n)) = n^2$. (ii) Both sides are Euler products, since Ψ_χ is completely multiplicative on ideals. At $\ell = 3$ both local factors are 1, because $\chi(3) = 0$. If $\ell \equiv 1 \pmod 3$, then ℓ splits into two primes of norm ℓ (Proposition 2.5) and $\chi_{-3}(\ell) = 1$, so both sides have local factor $(1 - \chi(\ell)\ell^{-s})^{-2}$. If $\ell \equiv 2 \pmod 3$, then ℓ is inert, $\mathrm{N}((\ell)) = \ell^2$ and $\chi_{-3}(\ell) = -1$, so both sides have local factor $(1 - \chi(\ell)^2\ell^{-2s})^{-1} = (1 - \chi(\ell)\ell^{-s})^{-1}(1 + \chi(\ell)\ell^{-s})^{-1}$. (iii) is proved in [26]. It agrees with the conductor–discriminant formula $3 \cdot \mathrm{N}\,\mathfrak{f}(\Psi_\chi) = \mathfrak{f}(\chi)\,\mathfrak{f}(\chi\chi_{-3})$, which gives $\mathrm{N}\,\mathfrak{f}(\Psi_\chi) = 3^{2K-1}$ for $K \ge 2$, since then $\chi\chi_{-3}$ is again primitive modulo 3^K. □

Remark 3.5 (No exceptional zeros for the moduli 3^K)**.** For real $\sigma > 0$,

$$L(\sigma, \chi_{-3}) = \sum_{m \geq 0} \big((3m+1)^{-\sigma} - (3m+2)^{-\sigma} \big) > 0,$$

since every term is positive; for instance $L(1, \chi_{-3}) = \pi/(3\sqrt{3})$. An exceptional (Siegel) zero of an L-function modulo q can only belong to a real non-principal character [8, Ch. 14]. By Corollary 3.3, the only such character modulo 3^K has the same L-function as χ_{-3}, which has no zero in $(0, 1)$. Hence no character modulo 3^K has an exceptional zero. This is Condition W3 of [26], which is proved there by exactly this argument. It implies that the prime number theorem for arithmetic progressions to the moduli 3^K holds in the Siegel–Walfisz range with effectively computable constants [8, Ch. 20]. It has no bearing on Conjecture 1.3: for $q = x^\theta$ one has $x^{1-c/\log q} = e^{-c/\theta} x$, so a zero-free region of width $c/\log q$ gives no saving for moduli of size a power of x.

4. Proof of Theorem 1.1

Proof of Theorem 1.1. Write $q = 3^K$, let $\gcd(a, 3) = 1$, and put $E(q, a) = \psi(x; q, a) - x/\varphi(q)$ and $y = (\log x)^{A+1}$.

Large moduli, $y < 3^K \leq x^\theta$. Since $\Lambda(n) \leq \log x$ for $n \leq x$, at most $x/q + 1$ integers $n \leq x$ lie in a given class modulo q, and $x/\varphi(q) = 3x/(2q)$, we have

$$|E(q, a)| \leq \Big(\frac{x}{q} + 1 \Big) \log x + \frac{3x}{2q} = \frac{x}{q} \Big(\log x + \frac{3}{2} \Big) + \log x.$$

The geometric series gives $\sum_{3^K > y} 3^{-K} < \frac{3}{2}\, y^{-1}$, and there are fewer than $\log x / \log 3$ values of K with $3^K \leq x^\theta$. Hence these moduli contribute

$$\ll \frac{x \log x}{y} + (\log x)^2 \ll \frac{x}{(\log x)^A}.$$

Small moduli, $3^K \leq y$. There are $O_A(\log \log x)$ of them. By the Siegel–Walfisz theorem [8, Ch. 22], $\max_a |E(q, a)| \ll_A x \exp(-c_A \sqrt{\log x})$ uniformly for $q \leq (\log x)^{A+1}$, so these moduli contribute $\ll_A x \exp(-c_A \sqrt{\log x}) \log \log x \ll_A x/(\log x)^A$.

Adding the two ranges proves the theorem. □

By Remark 3.5, the Siegel–Walfisz step, and hence the implied constant in Theorem 1.1, is effective for these moduli. For $\theta \geq 1$ the same argument applies, the only change being that each modulus $3^K > x$ contributes at most $\log x + 3/2$; this is the observation in Remark 1.2.

5. Where the value 5/8 came from (heuristic, not evidence)

This section explains why the programme expected the exponent $5/8$. It contains no argument for Conjecture 1.3.

(1) The value is $\theta_3 = (2^3 - 3)/2^3 = 5/8$ in the programme's formula $\theta_k = (2^k - k)/2^k$. That formula is a postulate of the programme (a system "of depth k" is defined to be one in which exactly k configuration classes are eliminated) [21, 24]; it is not a theorem. If "invalid" means "some constraint fails", then $2^k - 1$ of the 2^k configurations are invalid, not k.

(2) The depth $k = 3$ was assigned to cascade moduli through three "levels" modulo 3, 9 and 27. This assignment is proposed and currently unsupported. The $(1-\omega)$-adic filtration has $2K$ steps, not three (Remark 2.7). For twin primes $p > 3$, the conditions modulo 9 and 27 are implied by $p \equiv 2 \pmod 3$: the classes of p modulo 9 (or 27) allowed by $3 \nmid p(p+2)$ are exactly the lifts of 2 mod 3. By the programme's own independence criterion (C2) of [21], such coupled constraints give depth 1, for which the formula gives $\theta_1 = 1/2$.
(3) Pascadi's exponent 5/8 [13] concerns smooth numbers, and primes with triply-well-factorable weights, to general moduli. It is not about the moduli 3^K, and it neither confirms nor bounds anything in this paper. Its numerical agreement with θ_3 is a coincidence, not evidence.

Conjecture 1.3 lies above the GRH range 1/2, above the best known asymptotic range 5/12 for prime-power moduli, and above the range 0.4736 known for lower bounds (Remark 1.4).

6. The Hexagonal Lattice

This section records two elementary facts about the hexagonal lattice $\mathbb{Z}[\omega]$. Neither is used elsewhere in the paper.

Proposition 6.1 (Legendre duality of Eisenstein norm)**.** *The Legendre transform of the Eisenstein norm $f(a,b) = a^2 - ab + b^2$ is*

$$f^*(p,q) = \frac{p^2 + pq + q^2}{3},$$

with dual coordinates $p = 2a - b$, $q = 2b - a$.

Proof. $f^*(p,q) = \sup_{a,b}[pa + qb - f(a,b)]$; since f is a positive definite quadratic form, the supremum is attained at the critical point. Setting $\partial f/\partial a = p$ and $\partial f/\partial b = q$ gives $p = 2a - b$ and $q = 2b - a$. Inverting: $a = (2p+q)/3$, $b = (p+2q)/3$. Substituting back: $f^*(p,q) = (p^2 + pq + q^2)/3$. □

Remark 6.2. The factor 1/3 in f^* is $1/\det H$, where $H = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ is the Hessian of f. Here $\det H = 3$ is the absolute value of the discriminant -3 of the form $a^2 - ab + b^2$, which is also the discriminant of $\mathbb{Q}(\omega)$; so the 3 in the denominator is the same 3 that ramifies (Proposition 2.3). The original remark linking this factor to a "cyclic group enhancement" of Kloosterman sums is withdrawn.

Corollary 6.3 (Twin primes and splitting type)**.** *If $p > 3$ and $p + 2$ are both prime, then $p \equiv 2 \pmod 3$, so p is inert in $\mathbb{Z}[\omega]$, and $p + 2 \equiv 1 \pmod 3$ splits as $\pi\bar{\pi}$.*

Proof. Since $p > 3$ is prime, $3 \nmid p$. If $p \equiv 1 \pmod 3$, then $3 \mid p + 2$ and $p + 2 > 3$, so $p + 2$ is not prime. Hence $p \equiv 2 \pmod 3$ and $p + 2 \equiv 1 \pmod 3$; now apply Proposition 2.5. □

Remark 6.4. The splitting type of a prime $\ell \neq 3$ depends only on ℓ mod 3. Every prime $p \equiv 2 \pmod 3$ is inert, whether or not $p+2$ is prime, so the splitting type does not distinguish twin primes among the primes $p \equiv 2 \pmod 3$; Corollary 6.3 is a restatement of the congruence $p \equiv 2 \pmod 3$. The original statements that twin primes are "distinguished within $\mathbb{Z}[\omega]$ by their splitting type" and that $\theta_W = 5/8$ is "the level of distribution that this algebraic geometry forces" are withdrawn.

7. Withdrawn material

The original version proved Theorem 1.1 for $\theta = 5/8 - \varepsilon$ by an argument in two "geometric parts", followed by a uniqueness theorem and an appendix verifying "conditions of Watt's theorem". All of it is withdrawn. For readers of that version, we record where each step fails. Numbers such as "former Proposition 2.4" refer to the original version.

(1) *Three-level filtration (former Proposition 2.4).* There is no three-level structure; see Remark 2.7.

(2) *Triply-well-factorable weights (former Corollary 2.5).* Theorem 1.1 contains no sieve weights. The proposed pointwise factorisation $w(3^j) = \alpha(3^{j_1})\beta(3^{j_2})\gamma(3^{j_3})$ is not well defined, since $j = j_1 + j_2 + j_3$ in many ways. Weights are triply well-factorable, in the sense used by Maynard and by Pascadi [13], if for every factorisation $Q = Q_1Q_2Q_3$ they can be written as a Dirichlet convolution $\alpha * \beta * \gamma$ of 1-bounded sequences supported on $[1, Q_1]$, $[1, Q_2]$, $[1, Q_3]$: this is a property of a weight sequence over a dense set of moduli, not of a single modulus. The known results beyond level $1/2$ with factorable weights [6, 13] also concern a fixed residue class, whereas Theorem 1.1 has a maximum over a.

(3) *Type II bound and "ceiling $2/3$" (former Proposition 2.6 and Corollary 2.7).* The bound $E_{II} \ll x^{1/2}Q^{3/4+\varepsilon}$ was asserted without derivation: no bilinear form was opened, no completion step produced a Kloosterman sum, and sums over Dirichlet characters are not Kloosterman sums. Moreover, an error bound valid for all $Q \le x^{2/3-\varepsilon}$ would be a lower bound for the level, not a ceiling $\theta \le 2/3$.

(4) *Kloosterman sums (former Proposition 3.3).* The bound $|S(a, b; 3^K)| \ll (3^K)^{3/8+\varepsilon}$ is false. For even $K = 2m \ge 2$, writing each unit modulo 3^{2m} as $u(1 + 3^m t)$ and summing over t modulo 3^m gives

$$S(1, 1; 3^K) = 2 \cdot 3^{K/2} \cos(4\pi/3^K),$$

so $S(1, 1; 3^K) \ge 1.97 \cdot 3^{K/2}$ for even $K \ge 4$, and no bound $O(q^{1/2-\delta})$ with $\delta > 0$ can hold. Exact computation also gives $\max_{3\nmid b} |S(1, b; 3^K)| > 1.99 \cdot 3^{K/2}$ for $3 \le K \le 11$. The correct general bound is $|S(a, b; q)| \le \tau(q)\,(a, b, q)^{1/2} q^{1/2}$ [11, Ch. 11]. The gcd factor cannot be dropped, as the original statement $|S(a, b; 3^K)| \le 2(3^K)^{1/2}$ did: $S(0, 0; 3^K) = \varphi(3^K) > 2 \cdot 3^{K/2}$ for $K \ge 3$. The summed statement $\sum_{3^K \le Q} |S(a, b; 3^K)| \ll Q^{5/8+\varepsilon}$ is true, but it is weaker than what the general bound gives for fixed $(a, b) \ne (0, 0)$ (namely $\ll_{a,b} Q^{1/2+\varepsilon}$), so it is no improvement. Watt's theorem, invoked for the pointwise bound, does not give it (item (8)).

(5) *Exponent arithmetic.* Even granting the previous steps, the final inequality of the original proof,

$$x^{1/2}Q^{5/8+\varepsilon} \le x^{1-\varepsilon},$$

gives $Q \le x^{4/5-O(\varepsilon)}$, not $Q \le x^{5/8-\varepsilon}$; the exponent of Q was read off as if it were the level. A sharper estimate can only enlarge an admissible range, so the account "ceiling $2/3$ improved by $1/8$ to $5/8$" is incoherent; note also that $2/3 - 1/8 = 13/24$.

(6) *Vaughan's identity and Type I sums (former Section 4).* Vaughan's identity [17] is exact; it has no $O(n^{-1/2})$ term. With $U = V = x^{1/3}$ the Type I sums run over $m \le UV = x^{2/3}$, and the condition "$U < x^{1-\theta}$" used in the original version was not the relevant one.

(7) *Subconvexity.* Phragmén–Lindelöf interpolation gives the convexity bound, not a subconvexity bound. No subconvexity estimate was proved or used.

(8) *Watt's theorem and conditions W1–W3 (former Appendix A).* Watt's theorem [19] is a mean-value estimate in the t-aspect for $\int_0^T |\zeta(\frac{1}{2}+it)|^4 \left|\sum_{m\le M} a_m m^{-it}\right|^2 dt$. It involves no Dirichlet characters modulo q, has no conditions W1–W3, and gives no pointwise bound for a Kloosterman sum. The "verifications" were also invalid on their own terms: with $M = Q^{2/3}$ and $R = Q^{1/3}$, "W1" ($M \le 3^K$) fails for every $3^K < Q^{2/3}$, and "W2" ($MR \le 3^{2K}$) fails for every $3^K < Q^{1/2}$. "W3" conflated Siegel zeros with the bound $\theta \le 7/32$, which is the Kim–Sarnak bound [12] $\lambda_1 \ge 1/4 - (7/64)^2$ towards Selberg's eigenvalue conjecture, written in the normalisation $\theta = \sqrt{1-4\lambda_1}$; it is an unconditional theorem about Maass forms and is unrelated to zeros of Dirichlet L-functions. The Siegel-zero statement itself is true, for the reason given in Remark 3.5. The base change used for it does not have "matching" conductor (Proposition 3.4(iii)), and neither the class-number-one theorem of Baker [2] and Stark [15] nor Stark's effective Brauer–Siegel results [16] gives a zero-free region for L-functions of $\mathbb{Q}(\omega)$ of growing conductor.

(9) *Uniqueness and sharpness (former Theorem 5.1).* Its parts (1) and (2) rested on items (1)–(4) above. Part (3) is false as stated, since Theorem 1.1 holds for every $\theta < 1$. Pascadi's theorem [13] is an equidistribution result for a different problem, not a lower bound matching an upper bound, and no known result rules out a level above $5/8$ for these moduli. GRH gives level $1/2$ for a single modulus, not more than $5/8$.

(10) *Citations.* Polymath8 and Pascadi were misdescribed in the original introduction; the corrected descriptions are in Section 1. The paper "On the large sieve" is by Bombieri alone [5]; Vinogradov's contribution is [18].

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